

SEDAR 7: Gulf of Mexico Red Snapper

Data Workshop

Life History Group

- More movement information via tagging is needed from the western Gulf. There was discussion that a recreational tagging data base from the Coastal Conservation Association (CCA) may be available for this purpose. The sub-group recommended every effort be made to access and analyze this data base (by LSU researchers).
- The results from the otolith micro-chemical analysis were compelling in providing estimates of mixing rates for the north-central, northwest, and southwestern Gulf. The sub-group recommends continued work to also derive mixing rates from the eastern Gulf (west Florida shelf). It was of great interest to determine if there was evidence for localized recruitment in the east or whether recruits were derived from other areas as suggested by tagging results.
- Much more otolith microchemistry needs to be conducted on snapper off Texas, especially age 0 & 1 cohorts to aid in our understanding of the recruitment dynamics there.
- There needs to be an examination of whether regional stock recruitment functions can be developed. It was recognized that trawl surveys, which have been previously relied upon for recruitment estimates, are conducted from Texas to the Florida/Alabama border and may not capture any localized recruitment which may occur on the west Florida Shelf. The sub-group recommended that other survey methods be examined for recruitment determination and the red snapper larval index was recognized as a candidate for this purpose.

Shrimp Fishery Bycatch Group

- Future recommendations for improved data collection methods related to shrimp effort estimation include implementation of the Electronic Logbook Program (ELB) for 3-5 years (SEDAR7-REF-1; SEDAR7-REF-2) in conjunction with the current (or some form of) port agent interview system. Amendment 13 to the Shrimp Fishery Management Plan will address vessel monitoring systems (VMS) or ELB approaches for the shrimp fishery to obtain better effort data. Considerations of who will pay and own units (VMS or ELB) were discussed. VMS units are approximately \$1200 (+ monthly fee + maintenance) vs. ELB (\$500).

- The group strongly recommended a fully-funded shrimp trawl observer program to collect bycatch data as related to bycatch reporting requirements. This program would cost approximately \$2.5 KK annually.
- Work will continue on the new BRD designs using infrared observation technology (SEDAR7-DW-30). With this approach, we must encourage industry innovation by providing information to fishers for cooperative research to solve operational problems and maximize shrimp retention. The key to development of effective designs is getting new designs into the fleet, but this will result in innovation only if the industry has incentive to develop new technology. Consideration must also be given to the present certification protocol. BRD performance requirements will have to be re-examined based on performance projections of current BRD designs. BRD development should be focused on BRD designs which induce continuous and consistent bycatch escapement during variable environmental and commercial applications.

Assessment Workshop

- direct measurement of current fishing mortality rates,
- experiments to determine the magnitude and timing of density dependent compensation in juveniles,
- information on the effects on shrimp trawling on red snapper through community effects including nutrient cycling and changes in predation pressure,
- continuation and expansion of the fishery-independent survey for adult red snapper,
- more information on release mortality and discard rate by depth, season, and fishery,
- additional alternatives for reducing bycatch such as closed areas etc.,
- additional research such as simulation testing on the estimation properties of stock assessment methods and models,
- distribution and mixing between the East and West.

Review Workshop

Some of the following research recommendations are marked [D] or [A] or both. The symbol indicates that all or part of the corresponding recommendation was adapted from recommendations of the SEDAR 7 Data Workshop or Assessment Workshop.

- Data on shrimp fishery. The RW recognized the importance of obtaining better estimates of fishing effort in the shrimp fishery, which might be done through vessel monitoring systems, electronic logbooks, or otherwise [A]. Also, the RW recommends that the

statistical design and extent of the shrimp-trawl observer program be reviewed to ensure that the bycatch data collected are appropriate and sufficient for stock assessment.

- Independent estimates of mortality rates. Direct estimation of mortality rates through tagging would reduce uncertainty in future assessments [A].
- Fishing power. Research is recommended to estimate (independently of any stock assessment) changes in catchability q by gear over time. The RW believes that the introduction of GPS and marine chart-plotting equipment is likely to have increased fishing power substantially for some modes of fishing. Independent collection of data on fishing effort would provide valuable data for assessment and relieve the need to estimate catchability changes.
- Stock structure. Research (e.g., tagging, otolith analysis) is recommended to better describe stock structure and mixing rates. Research should include a review of oceanographic data to see whether transport from the Campeche Banks could reasonably be supplying important numbers of larvae to the western Gulf stock [A].
- Spawning-stock index. Given the many factors that can mask relationship of larvae to spawners, the value of the larval indices should be reviewed.
- Spatial distribution at age. The RW recommends study of the age structure observed from longlines (survey and fishery), to clarify geographic distribution of fish as they age.
- Density dependence. Research could clarify the magnitude and timing of density dependent compensation in juveniles by estimating survival (from age-0 to age-1 year) at different densities of juvenile abundance [A].
- Ecosystem concerns. The RW recommends that the management objectives for the fishery complex (shrimp, red snapper, vermilion snapper, etc.) be formalized. Simulation studies could usefully identify and evaluate appropriate management strategies (including use of various reference points) and corresponding assessment modeling approaches. Research could also test the hypothesis that red snapper production is enhanced in some way by increased shrimp trawling [A].
- Assessment modeling. The RW's recommendations for assessment modeling are made while recognizing that technology is currently limiting (the power of current small computers is marginal for the given model complexity). (a) Future assessments should include interval estimates on parameters and status indicators. (b) More diagnostic and output information should be provided in future assessment reports (e.g., plots or tables of F at age and plots of standardized residuals). (c) Extensive simulation tests of assessment models are recommended to examine accuracy, precision, and robustness [A].

- Age sampling. The RW recommends that representative sampling of age- and length-composition of red snapper be conducted consistently across area, time, and gear.
- Fecundity at age. The RW noted that few fecundity samples were available from older fish, and recommends that more such samples be collected.
- Model implementation. The RW recommends that the assessment model's recruitment sub-model be generalized to allow various options on the timing of bycatch mortality relative to density dependent compensation (see AW-8).

Recommendations of the CIE Contractors

CIE Chair

- Provide more clarity with regard to the exact role of the RW and the authority of the RW
- There needs to be a process for addressing potential disagreements between the RW and the AW and it must be made clear who has ownership of the Advisory Report.
- Supply all documents electronically with only essential reading provided as hardcopy.
- The red snapper assessment had not been updated since 1999. The DW, AWs, and RW to update the assessment have spanned more than a year. The whole process was delayed because of problems encountered with the previous assessment method when new data were added. Had a "simple" update been possible there would not have been the need for two AW's, and the full results would have been presented to an AW, rather than only becoming available at the RW. There is perhaps a lesson here. A simple update was not the objective of the first AW given the ambitions of the DW to produce and use an ultra-historical catch history. Simple updates can be done in a timely manner to provide appropriate advice to fisheries managers. However, with such a large gap between assessments, it was unlikely that a simple update would eventuate.
- In terms of providing timely scientific advice to fisheries managers, I have long advocated that there should be two asynchronous processes. Management advice should be provided by "simple" updates of stock assessments as required. The development of assessment methods and the substantial modification of data sets should be done in a separate process – it is harder and the timelines cannot be guaranteed. Scientific disagreements can also be dealt with outside of the management process.

CIE Reviewer

- Adequacy and appropriateness of data
 - * Perform sensitivity analyses to examine the effect of different historical catch divisions between east and west areas of the Gulf on the assessment.

- * Perform sensitivity analyses to examine the impact of potential changes in biological parameters over time on the assessment.
- * Examine the implications of the different potential distributions of larvae and adults for the assessment. Are there areas offshore suitable for juvenile settlement? Is the offshore age structure consistent with recruitment directly to deeper waters, or ontogenetic migration? Does oceanographic information suggest that larval movements of this type are realistic? Consider tagging programmes to examine the movement of juveniles and adults offshore/onshore and between east and west regions of the Gulf.
- * Consider the examination of available information on fishing position through logbooks (if sufficiently accurate) or observer programmes (if available) for serial depletion. Recommendations by the RW to examine the feasibility of VMS may need to be initiated before this can be investigated further.
- * Examine the sensitivity of assessment results to different values of release mortality rate (within the bounds indicated by the existing research). Investigate the interaction between natural mortality values and release mortality rates at younger ages.
- Adequacy, appropriateness and application of assessment methods
 - * The model represents a change from that applied during the 1999 assessment. Recommendations arise as result of this change, settings within the assessment, and particular assessment results:
 - * Examine the fishing mortality levels output from ASAP and CATCHEM for the short time period run to identify any differences and trends in this metric.
 - * Perform projections based upon the CATCHEM outputs from the short time period run to identify whether there are quantitative differences in expected recovery period. This will also require consideration of the management benchmarks resulting from changes in the estimated stock recruitment relationship, which may result in more significant differences.
 - * Consider the inclusion of migration between east and west areas of the Gulf in the model. Parameterization might be based upon available information (if sufficient) or through new tagging studies (if feasible).
 - * Examine the issue of density dependence and its effect on stock status and recovery further. Consider results in terms of risk to the population.

- * Present confidence limits on the recent recruitment levels estimated by the model, so that statistical differences between recruitments in the recent past and the ultra-historical period can be identified.
- * Develop further diagnostic approaches to assess the performance of the model. Present interval estimates for output parameters, or examine posterior distributions, as many of the estimates may be against their bounds (a count of the number of parameters against their bounds could be another diagnostic). Examine the shape of the response surface to assess whether local maxima are being identified. Perform retrospective analyses to assess model stability.
- Adequacy, appropriateness and application of population benchmark estimation methods
 - * Management benchmarks for these projections were highly sensitive to management decisions and biological assumptions. Recommendations are:
 - * Identify benchmarks that are more robust to changes in management levels and the stock-recruitment relationship, through management strategy evaluation simulations.
 - * Consider whether there is a need specifically to examine the red grouper/vermillion snapper fisheries (closed-season bycatch) along with the shrimp bycatch fishery and the targeted fisheries in assessments and management. Evaluate multispecies benchmarks.
 - * While the RW was not tasked to look at management issues, the division of the stock between east and west areas of the Gulf within the assessment allows separate management to be applied within these areas, rather than the current strategy of producing Gulf-wide management (TACs). Indeed, given that the eastern stock appears to be less productive than the western stock, Gulf-wide management has the potential to reduce the eastern stock to very low levels. This needs to be presented to managers for consideration.
- Adequacy, appropriateness and application of projection methods
 - * Consider performing stochastic projections and providing management with suitable diagnostics for recovery (e.g. the likelihood of recovery within particular time periods).